Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Morin, Lawrence P Ph.D.		
Project Title:	Circadian and Vestibular System Relationships		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	SUNY- The State University of New York		
PI Address 1:	Health Science Center		
PI Address 2:	Stony Brook University		
PI Web Page:			
City:	Stony Brook	**State:**	NY
Zip Code:	11794-8101	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	07/01/2001	**End Date:**	10/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-HPF00207		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This project sought to investigate the relationship between the vestibular system and circadian rhythmicity at a behavioral and anatomical level. The logic for the project was based upon suggestions that the medial vestibular nucleus (MVe) might connect with the intergeniculate leaflet (IGL) of the circadian system. Specific aims were devoted to optokinetic visual manipulations in an attempt to induce illusionary distortions of the vestibular system that might mimic non-photic stimuli known to modify circadian rhythm phase. These studies were unsuccessful. However, the project also addressed anatomical issues related to connectivity between the circadian and vestibular systems. These studies yielded three large paper and an extremely important result, namely, they demonstrate the convergence of three brain networks, hitherto considered substantially separate. In particular, the IGL has extremely widespread interconnections with much of the subcortical brain which are generally bilateral and reciprocal. Many of the areas with which the IGL is connected also receive vestibular input and/or project to the MVe. Finally, the data demonstrate that many of the regions connected		

	to both the MVe and IGL are also recipients of projections from the hypocretin-containing, sleep regulation system. It is likely that, in some fashion, the IGL contributes to the regulation of rapid eye movements during sleep.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The anatomical results have enormous implications for the study of sleep, vestibular and circadian systems and how they interact.
Task Progress:	We have completed anatomical studies demonstrating linkages between the sleep, circadian and vestibular systems. We have determined that it is unlikely that visual activation of the vestibular system alters circadian rhythm phase.
Bibliography Type:	Description: (Last Updated: 02/27/2004)
Articles in Peer-reviewed Journals	Horowitz, S. S., J. H. Blanchard, and L. P. Morin "Intergeniculate leaflet and ventral lateral geniculate nucleus afferent connections: an anatomical substrate for functional input from the vestibulo-visuomotor system. Journal of Comparative Neurology" 227-245 , Jan-2005
Articles in Peer-reviewed Journals	Horowitz, S. S., J. H. Blanchard, and L. P. Morin "Medial vestibular connections with the hypocretin (orexin) system. Journal of Comparative Neurology" 127-146 , Jan-2005
Articles in Peer-reviewed Journals	Morin, L. P. and J. H. Blanchard "Descending projections of the intergeniculate leaflet: relationship to the sleep/arousal and visuomotor systems. Journal of Comparative Neurology" 204-216 , Jan-2005
Presentation	Morin, L. P. "Seeing the brain through the circadian visual system. " Apr-2005